Spot a plant. Report it. Restore the area.

BioGuard helps staff photograph a plant, send a sighting and track the work needed to remove a confirmed invasive plant and replant the area.

Project focus: invasive-plant management in Sime Darby Property (SDP) green spaces.

Why do we need it?

Protect local plants

Invasive plants can spread and out-compete native plants for space and resources.

Protect habitats

Changes to plant life can reduce biodiversity and affect the wider ecosystem.

Connect the work

BioGuard links a sighting, an approved job, field photos and final review.

The complete flow

1. **GROUNDS** — **Take or choose a plant photo.**
   Start from the camera or gallery.
2. **BIOGUARD** — **Get a suggested plant identity.**
   See common name, scientific name and invasive-list status.
3. **GROUNDS** — **Send a sighting report.**
   Add the location and a short note.
4. **MANAGEMENT** — **Review and assign one restoration job.**
   Confirm the plant and invasive status; approve the work plan.
5. **ECOLOGY** — **Remove and replant as instructed.**
   Upload dated photos after removal and after replanting.
6. **MANAGEMENT** — **Check the evidence and close the ticket.**
   Ask for changes if the work needs correction.

One job covers removal and replanting. If planting is delayed, record the reason. The same job stays active until planting is finished and management approves it.

Three roles. One shared sign-in.

Each account opens its own workspace automatically.

GROUNDS

Report sightings

Photograph a plant, send its location and notes, and follow the report status.

ECOLOGY

Complete assigned work

Accept jobs, follow the approved instructions, and send removal and replanting evidence.

MANAGEMENT

Review and coordinate

Confirm reports, assign the ecology team, check evidence, and close completed tickets.

What the AI can do

The model suggests a plant. A person makes the decision.

The current model is EfficientNetV2B0, trained using a dataset of **2,700 iNaturalist photos across 9 plant species**. It recognises this limited set; it cannot identify every plant.

It achieved about **86% accuracy on 380 historical test photos**. That does not guarantee the same result on new phone photos. A confident result can still be wrong.

Not confident means the suggested match needs extra care. Invasive status follows the project reference list; management confirms it before work is assigned.

What your team should know

A complete job needs two evidence photos.

One after removal and one after replanting. Completed work goes to management for review before it is closed.

Reporting is a deliberate action.

Identification alone is temporary. Submitted reports and evidence are saved; unfinished drafts stay on the device.

The beta needs the laptop server.

The Android APK and Expo app connect to the server for identification and reports. Remote testing needs the laptop online.

First team test: sign in -> identify a photo -> send a clearly labelled test sighting -> assign a restoration job -> upload test evidence -> review and close.

Environmental background: Convention on Biological Diversity - Invasive Alien Species.